Mobile communication using a mobile phone with a glucometer for glucose control in Type 2 patients with diabetes: as effective as an Internet-based glucose monitoring system

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Summary

A mobile phone with a glucometer integrated into the battery pack (the 'Diabetes Phone') was launched in Korea in 2003. We compared its effect on management of type 2 diabetes to the Internet-based glucose monitoring system (IBGMS), which had been studied previously. We conducted a randomized trial involving 69 patients for three months. Participants were assigned to an Internet group or a phone group. The phone group communicated with medical staff through the mobile phone only. Their glucose-monitoring data were automatically transferred to individual, web-based charts and they received medical recommendations by short message service. The Internet group used the IBGMS. There were no significant differences between the groups at baseline. After three months' intervention, HbA_{1c} levels of both groups had decreased significantly, from 7.6% to 6.9% for the Internet group and from 8.3% to 7.1% for the phone group ($P < 0.01$). Levels of patient satisfaction and adherence to medical advice were similar. Mobile, bidirectional communication between doctors and patients using the diabetes phone was as effective for glucose control as the previously-studied Internet-based monitoring system and it was good for patient satisfaction and adherence.

Introduction

Various strategies using electronic technologies or educational programmes have been proposed to improve the quality and efficiency of care for people with diabetes.[1–11] In a previous study, we introduced a new bidirectional communication tool for diabetes management, termed the Internet-based glucose monitoring system (IBGMS), and demonstrated its short-term and long-term effects on glycaemic control in patients with type 2 diabetes.[12,13] Other researchers have also shown the effectiveness of telecommunication between health-care providers and patients for maintaining appropriate blood glucose control.[14–17] These systems allow patients to send data to doctors, identify specific problems and receive recommendations from health-care providers more frequently and with fewer time and place limitations than at traditional face-to-face outpatient clinics. With the IBGMS, doctors can check patients' self-monitored blood glucose data at regular intervals, and patients can communicate with their doctors whenever they have questions. Although bidirectional interactive communications between doctors and patients can be achieved with the IBGMS, there remain time and place limitations. Patients must connect to the Internet using their personal computer to input their data or questions on the IBGMS, while doctors only visit the website at regular intervals unless they have received questions from patients.

To improve the efficiency of communication, we linked the IBGMS to the short message service (SMS) of mobile phones.[18] The patients sent their data to doctors through the IBGMS and then received the doctors' recommendations about adjustment of drug dosage and other related data on a mobile phone. This system improved the efficiency of communication and patients' satisfaction also increased slightly. However, patients were still limited by place and time when sending questions and blood glucose-monitoring data.

Recently, mobile phones have been developed that can measure blood glucose level and transmit data directly to

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a web server (e.g. the Diabetes Phone – see Figure 1). The combination of a glucose meter with a mobile phone makes it easier for patients to report their self-monitored blood glucose (SMBG) levels and receive feedback from their doctors. We have investigated the effectiveness of the diabetes phone for blood glucose control and compared it with the effectiveness of the IBGMS.

Methods

Study subjects were recruited from the outpatient clinic of the Diabetes Centre in Kangnam St Mary's Hospital between January and April 2005. The inclusion criteria were patients with type 2 diabetes mellitus who were able to access the Internet and to communicate through a mobile phone using the SMS. Criteria for exclusion included significant disease conditions such as symptomatic heart failure, aspartate aminotransferase (AST) and alanine aminotransferase (ALT) levels more than twice the normal level, renal disease and creatinine level greater than 1.5 mg/dL. Patients were also excluded if they participated in other programmes that provided information or education about diabetes management from any Internet websites or by any mobile phones other than ours. All patients were interviewed initially and gave informed consent. A total of 75 patients with type 2 diabetes mellitus were enrolled voluntarily and 69 patients completed the three-month protocol.

Device

We distributed diabetes phones (Mobile Health-care diabetes phone; LG-KP8400, LG, Seoul, Korea) to patients in the phone group. The diabetes phone is a mobile phone containing a device to measure capillary blood glucose on

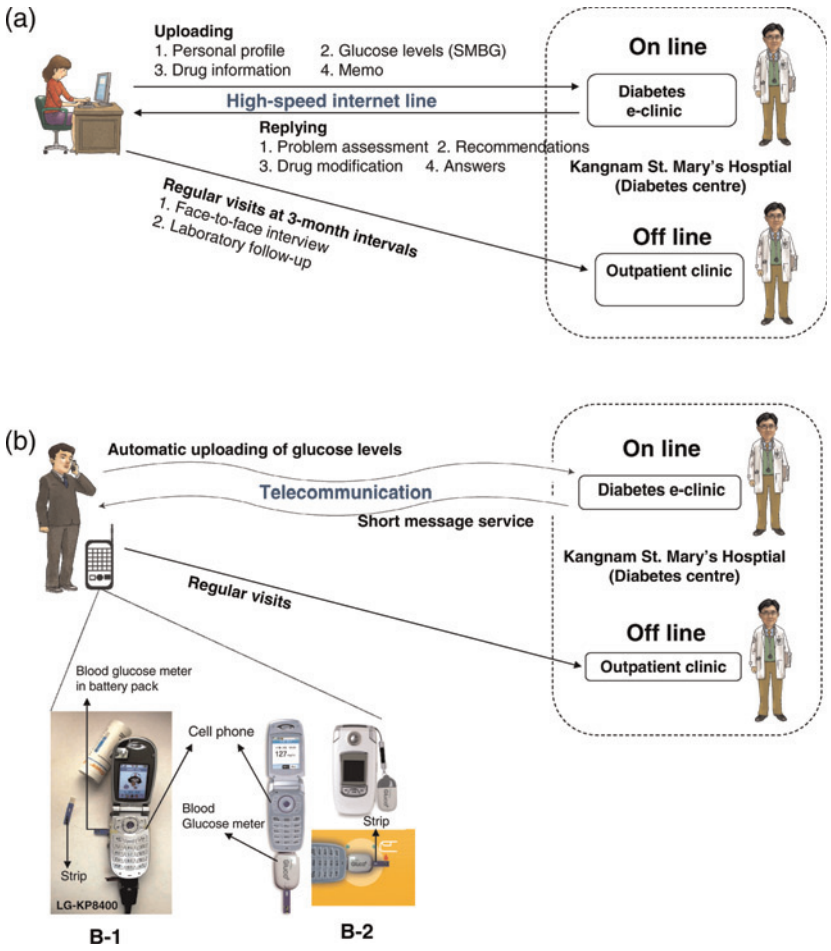

Figure 1 (a) The Internet-based glucose monitoring system (IBGMS). Patients logged on to the website from their homes or offices at a time they found convenient and uploaded their glucose data, together with additional information such as current drug information (the type and dosage of oral hypoglycaemic medications or insulin), lifestyle modifications and hypoglycaemic events. In addition, patients recorded any changes in their blood pressure or weight, and any questions or detailed information that they wished to discuss, such as changes in diet, exercise, hypoglycaemic events, and other factors that might influence their blood glucose level. (b) The telecommunication-based glucose control system in which mobile devices such as a mobile phone with the capacity to measure blood glucose were adopted. Patients could upload their information through telecommunication automatically and patient–doctor communication could be conducted in real time by the short message service. B-1 shows the first generation of diabetes phone (LG-KP8400) and B-2 shows the second generation of diabetes phone (Gluco Plus)

site and transmit blood glucose data to a web server automatically without manual input (Figure 1b-1). The diabetes phone has the same functions as other mobile phones, including SMS and an alarm. Patients in the IBGMS group were given glucose meters (Accu-Check Active, Roche, Co.) for measuring SMBG levels.

Study design and intervention

The study was a randomized, non-inferiority study with an active-controlled period. After permission from the appropriate ethics committee, subjects were enrolled for the study in the outpatients' clinic. Participants were randomly assigned to either the Internet ($n = 37$) or phone ($n = 38$) group. Weight, height, waist circumference and blood pressure were measured at the initial and final interview. Blood and urine samples were collected at the first visit and at the final visit, three months after enrollment, for measurement of fasting blood glucose, two-hour postprandial glucose, lipid profiles, HbA_{1c}, AST, ALT, blood urea nitrogen and creatinine. HbA_{1c} was measured by high-performance liquid chromatography (Variant II HbA_{1c} analyzer; Bio-Rad, Montreal, Quebec, Canada).

To ensure all patients understood self-management of diabetes mellitus, an education team consisting of an endocrinologist, a certified nurse educator and a dietician provided a diabetes education programme for all participants. The patients were also taught how to perform their SMBG measurements and were given information about diet, physical exercise and managing hypoglycaemia. Individual patients were taught about the frequency and timing of SMBG measurements according to the mode of treatment and degree of glycaemic control. We recommended that SMBG measurement be performed at least once a day in all patients, even after they had achieved target blood glucose levels. We recommended that patients being treated with insulin should measure their SMBG level three times a day. For patients being treated with oral agents who had not achieved target HbA_{1c} levels, we recommended SMBG measurement at least twice a day. Participants in the Internet group were taught about accessing and using the specialized, web-based diabetes patient management system and how to communicate with a management team through their individualized, web-based charts on the Internet website,[13] at least once every other week. However, they could ask personal questions whenever these arose and a doctor replied within two working days. The patients in the Internet group had to enter their self-monitored glucose data on individual web charts, while data from patients in the phone group were automatically transmitted to a web server. One doctor reviewed all the information about SMBG data on individual web-based charts at least once every other week and sent recommendations to the Internet group through the Internet web chart and to the phone group by SMS on the diabetes phone. The systems are illustrated in Figure 1a and 1b. Patients in both groups received visual displays with graphs showing blood glucose levels and the degree of glucose control for the previous 24 hours, one week and one month. All patients received recommendations from a doctor through the Internet or by phone once every other week. Patients in either group who did not submit blood glucose levels for longer than one week received a simple message encouraging SMBG measurement and asking them to enter their data. Patients who did not transmit their glucose levels for three consecutive weeks were withdrawn from the study.

Satisfaction and adherence

To examine the degree of satisfaction with the medical service and adherence with the doctor's recommendations, questionnaires were used at the end of the study. The patients' answers were based on self-report. To assess the degree of satisfaction and adherence, we classified them into five and four levels, respectively. For satisfaction, level I was more than 90% satisfaction; II, more than 80% satisfaction; III, more than 50% satisfaction; IV, more than 30% satisfaction; and V, more than 20% satisfaction. For adherence, level I meant that patients obeyed more than 90% of the doctor's instructions; level II, 70–90%; level III, 50–70%; and level IV, less than 50%. Thus, Level I and II represented good levels of satisfaction and adherence.

Statistical analysis

The data were analyzed using the SAS program (version 6.12, SAS institute, Cary, North Carolina). A chi-squared test and unpaired t-test were used to evaluate the homogeneity of demographic and clinical characteristics between the Internet and phone groups. A paired t-test was used to assess the difference of the variables between baseline and follow up in both groups. Fisher's exact test was used to test for difference in satisfaction with care and adherence to doctors' orders.

Results

Characteristics of patients

110 patients expressed their intention to participate in the study; 35 patients were excluded. Twenty-seven patients were excluded due to significant diseases that could affect the study outcome. Eight patients refused to participate in our study after learning that the study required frequent access to our homepage and frequent self blood glucose monitoring. Of the 75 subjects who volunteered to participate, 69 (92%) patients completed the study. Dropout rates were similar between groups, with three subjects in each group not completing the study. Two patients were withdrawn because of missing laboratory follow-up tests while four patients were withdrawn when they did not send

Table 1 Clinical characteristics and baseline laboratory data for IBGMS and phone groups. Mean (SD) values shown

Characteristics	Internet group (n = 34)	Phone group (n = 35)	P
Age, years (SD)	45.2 (11.3)	51.1 (13.2)	0.11
Male/female	26/8	28/7	0.31
Body mass index, kg/m^2 (SD)	23.6 (3.0)	25.3 (4.7)	0.06
Diabetes duration, years (SD)	5.3 (4.8)	8.2 (7.8)	0.43
Systolic blood pressure, mmHg (SD)	126.4 (12.8)	131.1 (13.4)	0.17
Diastolic blood pressure, mmHg (SD)	79.5 (10.4)	79.3 (10.9)	0.69
HbA_{1c}, % (SD)	7.6 (1.9)	8.3 (2.3)	0.30
Fasting plasma glucose, mg/dL (SD)	167.5 (73.7)	146.6 (49.7)	0.14
Two-hour postmeal glucose, mg/dL (SD)	269.2 (99.1)	246.1 (65.0)	0.26
Total cholesterol, mg/dL (SD)	175.0 (38.3)	187.5 (28.6)	0.13
Triglycerides, log mg/dL (SD)	4.4 (1.0)	4.7 (0.7)	0.11
HDL cholesterol, mg/dL (SD)	50.8 (14.7)	47.3 (14.4)	0.37
LDL cholesterol, mg/dL (SD)	97.0 (26.9)	109.0 (29.1)	0.055

HDL, High-density lipoprotein; LDL, Low-density lipoprotein

blood glucose data for more than three weeks during the study period. The groups did not differ significantly in age, body mass index, diabetes duration, systolic and diastolic blood pressure, HbA_{1c}, fasting and two-hour postprandial glucose and lipid profiles (Table 1). Most participants were male: 76% of the Internet group and 80% of the phone group.

Frequency of glucose monitoring

In total, 6287 blood glucose measurements for the Internet group and 6487 measurements for the phone group were transmitted to a web server during the study. The mean number of days on which patients measured their own glucose at least once a day was 77 (SD 16) in the Internet group and 84 (SD 29) in the phone group ($P = 0.25$). The mean frequency of SMBG measurements per day in all the patients was 2.3 (SD 1.0), and there was no difference between the Internet and phone group: 2.1 (SD 0.7) and 2.4 (SD 1.3), $P = 0.30$.

Effects of intervention

The baseline and follow-up glucose and HbA_{1c} data after three months' intervention are shown in Figure 2. The basal HbA_{1c} values were 7.6% (SD 1.9) in the Internet group and 8.3% (SD 2.3) in the phone group ($P > 0.05$). HbA_{1c} decreased significantly in both groups after three months' intervention ($P = 0.001$), and the mean HbA_{1c} decrement between the two groups was not different significantly ($P = 0.27$). Two-hour postprandial glucose levels also decreased significantly in both groups after three months ($P = 0.001$), but fasting plasma glucose levels did not change ($P = 0.07$). The changes in mean fasting plasma glucose levels (Internet vs. phone: 167.5 to 148.8 mg/dL vs. 146.6 to 143.4 mg/dL, $P = 0.33$) and two-hour postprandial glucose levels

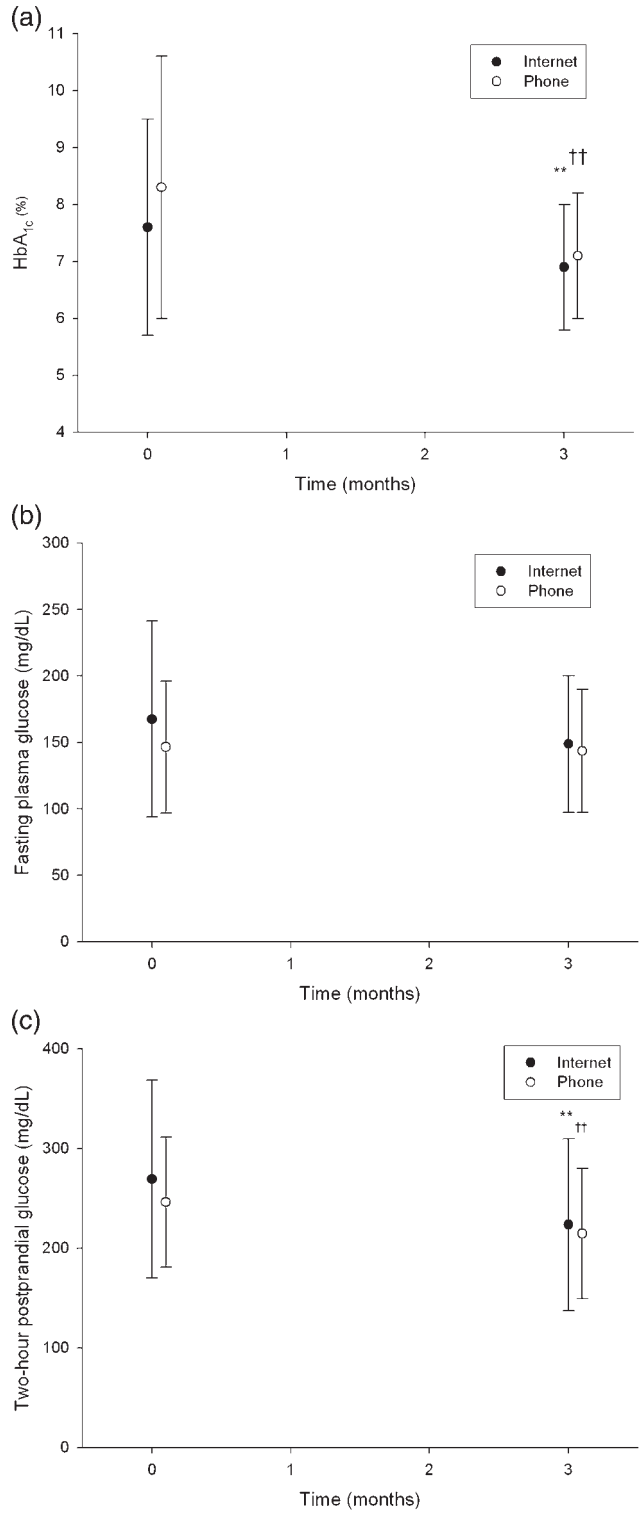

Figure 2 (a) Changes in HbA_{1c} levels in the two groups. The changes from baseline to 3-month follow up were significant ($P < 0.01$) in the Internet (**) and phone (††) groups; (b) Changes in fasting plasma glucose levels in the two groups; (c) Changes in two-hour postprandial glucose levels in the two groups. The changes from baseline to 3-month follow up were significant ($P < 0.01$) in the Internet (**) and phone (††) groups

(Internet vs. phone: 269.2 to 223.5 mg/dL vs. 246.1 to 214.6 mg/dL, $P = 0.48$) during the study period were not significantly different between the two groups.

Table 2a Difference in satisfaction with medical service to control diabetes

n (%)	I	II	III	IV	V	*P*
Internet group	9 (29)	16 (52)	5 (16)	1 (3)	0 (0)	0.94
Phone group	10 (35)	13 (45)	5 (17)	0 (0)	1 (3)	

I. more than 90% satisfaction; II. more than 80% satisfaction; III. more than 50% satisfaction; IV. more than 30% satisfaction; V. more than 20% satisfaction

There were no changes in total cholesterol, HDL cholesterol, triglyceride and LDL cholesterol levels before and after three months' intervention within and between groups (data not shown).

Both groups showed similar levels of satisfaction with the service. There were good levels of satisfaction in both groups, with no difference between the two groups (Internet vs. phone: 81% vs. 79%, $P = 0.94$) (Table 2a). For the questions about adherence with the doctor's recommendations, more than 70% of subjects replied that they accomplished the tasks as recommended by the doctor (Internet vs. phone: 77% vs. 83%, $P = 0.999$) (Table 2b).

Discussion

In a previous study, we reported that bidirectional communication between patients and doctors using the Internet was effective for glucose control in type 2 diabetes in the long-term as well as the short term.[12,13] The present study first demonstrated that telecommunication using a mobile phone with a glucometer was also effective for glucose control in a short term period of 3 months, as its effect was similar to that of the IBGMS. Baseline HbA_{1c} levels were significantly decreased in both groups after 12 weeks' intervention. Although the decrease in HbA_{1c} levels was slightly greater in the phone group (phone group vs. IBGMS group: −1.2% vs. −0.7%, respectively), there was no significant difference in HbA_{1c} levels between the two groups at the end of the study. The fasting and two-hour postprandial blood glucose levels of the two groups showed the same pattern as HbA_{1c}. These results provide good evidence that the diabetes phone might be useful for self-management in patients with type 2 diabetes. It might be expected that the greater mobility and easier access of the diabetes phone, compared to the IBGMS, would lead to more beneficial effects, but we did not observe this in the present short-term study. This was unsurprising because of the previously demonstrated, potent, short-term intervention effects of the IBGMS.

Table 2b Difference in adherence to doctors' advice

n (%)	I	II	III	IV	*P*
Internet group	8 (26)	16 (52)	6 (19)	1 (3)	0.999
Phone group	7 (24)	17 (59)	5 (17)	0 (0)	

I. patients obeyed more than 90% of the doctor's recommendations
II. patients obeyed 70–90% of the doctor's recommendations
III. patients obeyed 50–70% of the doctor's recommendations
IV. patients obeyed less than 50% of the doctor's recommendations

In a recently published randomized controlled trial, the effect on glycaemic control of a glucometer connected to a mobile phone with support by a nurse was tested in young adults with type 1 diabetes. This study demonstrated a decrease of HbA_{1c} in the diabetes phone group (intervention group) over time.[14] There was, however, no significant difference in HbA_{1c} between the intervention and control groups, probably because of disease severity and low motivation. In contrast, some studies reported that the mobile phone was available for glucose control in adolescents with type 1 diabetes and adults with type 2 diabetes.[1,2,11] In the present study, we have shown that intervention using the diabetes phone decreased HbA_{1c} levels as much as the IBGMS, which already showed a superior effect over a control group. The result may have been due to the synergistic effects of mild disease, adequate guidance by a doctor and enthusiasm of patients in caring for their diabetes. We recruited volunteers for our study and almost all the subjects were highly motivated to control their blood glucose levels. Furthermore, all eligible patients had to be familiar with handling the Internet and SMS function of the mobile phone. As a consequence, most of our subjects were male, less than 55 years old and with high levels of education. These factors might have contributed to the better glycaemic control than shown in other studies. The visual displays of glucose profiles were sent to the phone group, but they were limited compared to the detailed information provided on the web pages to the IBGMS group. However, our results showed that this did not affect the degree of improvement in glycaemic control for the phone group.

When the diabetes phone group was compared with the Internet group, there were no differences in degree of satisfaction about the medical service or in glucose control. As we reported before, patients using the SMS in addition to the IBGMS for their diabetes management showed slightly improved patient satisfaction levels.[18] In the current study, satisfaction levels in both groups were similar and all patients were satisfied with the medical service provided through the new device. Unfortunately, it was not possible for patients to compare which method was better in this study, because participants in one group did not experience the other group's method. In addition, assessment of satisfaction and adherence was based on self-reporting, which might limit the accuracy of the result. With regard to satisfaction, the phone group enjoyed greater convenience in terms of mobility and real-time communication than the Internet-only group. However, the phone group also had complaints about the brevity of the recommendations sent through the SMS function. Therefore, a combined system of telecommunication using the diabetes phone and the Internet-based system would probably be more effective.

While we observed positive results for the diabetes phone in the present study, there are several obstacles that need to be overcome before this system could be implemented

widely. The high cost of the diabetes phone was a major barrier to clinical implementation at the beginning of the study. Fortunately, 'GlucoPlus', an adaptor for glucose measurement used with a standard mobile phone, was developed during the study period (Figure 1b-2). The cost of the 'GlucoPlus' adaptor is similar to the usual glucometer and it is small enough to carry around with a mobile phone. This development might accelerate the clinical implementation of the diabetes phone in future. Another important barrier is real-time communication with patients. In our system, quick and flexible modifications of lifestyle and medication dosage were not possible because of the doctor's workload. This led to the decreased satisfaction reported by some patients in the survey. Furthermore, it was impossible to respond promptly to patients' urgent reports of acute hypoglycaemia, symptoms of hyperglycaemia and so on. This limitation might be overcome by the development of real-time alarm software and autonomic response algorithms based on the patients' SMBG data.

In summary, our results suggest that the telecommunication system using the diabetes phone with a glucometer can be effective for glucose control and good for patient satisfaction and adherence.

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